FINITE LOCALLY FREE EQUALS FINITE FLAT AND LOCALLY OF FINITE PRESENTATION, AND FINITE FLAT OVER A LOCALLY NOETHERIAN BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement has two enumerations, the second carrying the label (3) for its single item; both enumerations are inside the counted statement span.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Locality of integrality, finite local freeness, flatness and local finite presentation for algebraic spaces. Lemmas lemma-integral-local, lemma-finite-locally-free-local, lemma-flat-local and lemma-finite-presentation-local, used at proof lines 9696-9702. Reduce every one of the three conditions to the base change V -> Y.
- Representability of the morphisms involved. used at proof line 9696. Makes the base change check legitimate for all three conditions.
- Local Noetherianness passes to an etale cover. used at proof lines 9703-9704. Puts the scheme case in the right setting for condition (3).
- The scheme case of the equivalence. Morphisms, Lemma morphisms-lemma-finite-flat, used at proof lines 9704-9706. Is the input that closes the argument.

Source reference. Original lemma 03ZW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5235–5235. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0410: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5275–5275. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03MM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4754–4754. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03ZQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5229–5229. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02KB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3366–3366. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The following are equivalent:*

(1) *f is finite locally free,*
(2) *f is finite, flat, and locally of finite presentation.*

If Y is locally Noetherian these are also equivalent to

(3) *f is finite and flat.*

Proof. In each of the three cases the morphism is representable and you can check the property after base change by a surjective étale morphism $V \to Y$, see Lemmas 03ZQ, 03ZW, 03MM, and 0410. If Y is locally Noetherian, then V is locally Noetherian. Hence the result follows from the corresponding result in the schemes case, see Morphisms, Lemma 02KB. □